Experimental design and independent acceptance criteria

	Structured workflow	Independent checkers	Cross-stage repair	Modelling retrieval	Clarification web lookup
Direct LLM	○	○	○	○	○
Structured workflow	●	○	○	○	●
Workflow with checkers	●	●	○	○	●
Complete workflow	●	●	●	○	●
Workflow with retrieval	●	●	●	●	●
Without clarification lookup	●	●	●	○	○

Every configuration searches for data; the final row disables only clarification lookup.

Five development families, two scales each

Taipei chargers
Population coverage

YouBike tour
Closed inspection tour

AED teams
Balanced partition

London ambulance
Facility assignment

Berlin playgrounds
New area coverage

External judge	Data correctness	Model correctness	Usable delivery

Frozen code, prompts and truth; independent outcomes; failures and unverifiable models retained.